Electric Transportation

Action Plan for India

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***Abstract*—After almost 70-years after independence in 1947, India stands at the deterministic crossroads of economic growth and energy needs for over a billion people, while facing challenges to lower carbon emissions and to improve energy security. India's transportation sector growth is one of the major contributors of deteriorating air quality in the urban cities, accounting for 30% of the total fine particulate matter ($PM_{2.5}$) emissions. India is taking significant policy initiatives to leapfrog and transform electric vehicle (EV) adoption to embrace a tryst with electric transportation. India's National Electricity Mobility Mission (NEMM) Plan 2020 outlines an ambitious target to deploy six to seven million hybrid and EVs. Through stakeholder feedback and qualitative and quantitative analysis, this paper reviews goals and objectives of India's policies, their associated challenges experienced by the Indian ministries, and suggests the action plan, which is unique to the Indian context, to accelerate EVs and EV charging infrastructure, and technology development and deployment to meet the NEMM targets. This first-of-its-kind analyses for India can accelerate EV adoption, open innovation opportunities for both transportation and electric industry, and support energy security and improved urban air-quality objectives.**

Keywords—India's policies, clean transportation, electric vehicles, charging infrastructure, technolgies and standards.

I. Introduction and Background

Diffusion of zero-emission electric transportation in India serves the objectives of lowering carbon emissions and improved energy security. As a relative case, in the United States (U.S) the transportation sector uses 28% of the total energy, mostly from petroleum-based sources [1]. In 2012, this sector contributed 28% of the total greenhouse gas (GHG) emissions [2]. To address the resulting climate-change impacts, the U.S. is undertaking many federal- and state-level initiatives for zero-emission electric vehicles [3][4]. While internal combustion engine (ICE) vehicles rely on fossil-based fuels (petroleum and diesel) to generate energy, electric vehicles (EV) rely on electricity to power the vehicle's electric motors. This transition from traditional ICE-based vehicles to EVs requires adequate electricity infrastructure and policies that support optimal and planned deployments [5][6]. Advancing zero-emission EVs for improved air quality and human health are and best promoted by the policies. Additional benefits of clean transportation are similar to the use of alternate fuels (e.g., hydrogen) or improvements in the fuel economy.

After almost 70-years after independence in 1947, India stands at the deterministic crossroads of economic growth and energy needs for over a billion people, while facing challenges to improve energy security and to lower carbon emissions, which also improves urban air-quality. With emphasis to the challenges faced by the growth in transportation sector, India's oil-independence, better urban air quality, and stronger economic development future relies heavily on aggressive adoption clean transportation policies. India imports more than 80% of its crude oil, but has one of the lowest vehicle to owner ratios: one vehicle for every 1,000 people [7]. Even with low adoption rates, the transportation sector is a major contributor to the deteriorating air quality in India's urban cities. The near-surface fine particulate matter ($PM_{2.5}$) levels experienced by the daily-commuters have an annual-mean of 15 to 20 times higher (~60 times in winter) than the World Health Organization's guideline of 10μg per-m3 for PM2.5 and 20μg per-m3 for PM10 [8][9]. In India's capital, Delhi, 30% of the total $PM_{2.5}$ emissions are from transportation [10]. In an urban infrastructure, emissions from ICE vehicles worsen air quality and public health [11][12]. With India's transportation sector is projected to add more than 250 million cars, 185 million two- and three-wheelers, and 30 million trucks and vans by 2040, these ownership rates will further strain oil supply imports and exacerbates the air quality [13].

In a progressive effort to improve the energy-security and quality of life of its people, the Indian government has unveiled the National Mission for Electric Mobility (NMEM) to introduce Hybrid and EVs [14]. The NMEM has an ambitious goals and an opportunity to deploy six to seven million hybrid and EVs, such as 2, and 3 wheelers, and light, medium, and heavy duty vehicles (e.g., cars, buses). While the progress of these policy decisions shall reveal over time, to support the NMEM goals and to accelerate the adoption of electric transportation in India, the study has following key objectives:

- Showcase key policy interventions for EV adoption.
- Identify challenges and propose specific actions.
- Propose charging station infrastructure solutions.
- Identify and propose electricity payment mechanisms.

The represents opportunistic analyses in support of the objectives to accelerate EV deployments, while considering the unique Indian challenges and providing an action plan and concluding recommendations. The initial analyses, which are under consideration by the Indian ministries, are representative priorities to initiate the EV adoption campaign. This study recommends policy and technology action plan to leapfrog EV deployments, and identifies priority sectors to address EV adoption challenges and optimal ranges (e.g., battery size) per single charge for different vehicle categories, reviews and recommends charging station levels and ownership models, and provides a framework for remote management of charging stations and electricity payment settlement mechanisms.

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II. Analyses of Policy and Technology Interventions

Under the aegis of the NEMM policies, the government of India has also launched the Faster Adoption and Manufacturing of Electric Vehicles (FAME) India scheme with a financial support to the tune of INR 800 crores (approximately USD 125 million) for the first two years, beginning April 2015. In support of the encouraging local growth, innovation, and economic development in the transportation sector, the FAME guidelines has four objectives [15]: (1) to meet the NEMM targets, (2) strengthen technological capabilities of EV and infrastructure, (3) set-up EV charging infrastructure, and (4) initiate field-studies to build consumer confidence and scale adoption. In planning the clean transportation rollout, one key element is availability of adequate charging stations to be set-up by the electricity distribution companies (DISCOM) and/or third-party agencies. The DISCOMs and/or third-party agencies need to be actively engaged under the NEMM to build charging infrastructure and future proof it for the EV diffusion.

The ideas and intrinsic process of diffusion (scaled adoption) enhances innovation. Without diffusion, innovation would have minimal social and economical impact. Innovation is also driven by the field results. Electric transportation policy objectives must support this process to drive innovation and to identify new technology inventions. For example, EVs enable Smart Grid users to be business-to-consumers, business-to-businesses, or machine-to-machine interactions.

Electricity rate tariffs for EV charging may be viewed from a different perspective, considering these are distributed energy resources (DER) for the electric grid. The figure below shows the role of EVs, as a DER, in this context and how they interact with the charging systems and the electric grid domains.

Fig. 1. Electric Vehcile Interfaces with Charging Stations and Electric Grid

In India, per existing laws, a distribution license is needed to sell electricity [16]. While this enables the DISCOMs to operate the EV charging stations, a third-party charging station owner or operator cannot buy electricity from a DISCOM and sell to an EV driver. This should ideally be addressed at the Electricity Act. But an Amendment to the Electricity Act is in the Indian parliament, which may be passed in the near future. Although electricity resale may be considered a commercial use, a separate tariff for EVs may be introduced that promotes faster EV adoption and enhances grid reliability [17]. Such tariffs for EVs must have time of the day or real time pricing such that, whenever there is surplus generation, the EVs can charge from the grid (e.g., smart or managed charging) and whenever there is generation deficit, the EVs should sell power to the grid, which is possible with Vehicle to Grid (V2G) technologies [18].

In addition to setting-up of adequate EV charging infrastructure and enabling consumer- and grid-friendly electricity rate tariffs, one key barrier for the EV adoption is the marginally higher initial costs, relative to ICE vehicles. The Indian government is working on schemes to encourage EV ownership like giving subsidy to end users for purchasing Electric Vehicle. EV battery, which mostly relies on Lithium-ion technologies, is a key component that adds to the EV cost [19]. Due to minimal components and lower fuel costs, relative to ICE, the total cost of ownership (TCO) can be lower. To further lower the TCO, the batteries at the end of its useful life for EV use, typically has >70% capacity. These batteries can be used for stationery applications such as solar, off-grid street lighting, peak hours, and to balance grid with variable renewable generation for several more years. The government's policies and education will play a key role in influencing such opportunities.

III. Challenges and Suggested Action Plan to Accelerate Electric Transportion

On the behest of the Indian Ministry of Power (MoP), the India Smart Grid Forum (ISGF) organization supported this study to aggregate various stakeholders from different domains and conduct workshops that provided feedbacks to identify challenges to accelerate the EV adoption in India, and support of the NEMM and FAME targets. Under India's smart cities mission, India's Ministry of Urban Development (MoUD) has selected twenty cities, as "*Lighthouse Cities*" through a national competition. Out of these twenty lighthouse cities, twelve of them have proposed EVs deployment, as a priority. These twelve Indian cities are: Belagavi, Bhubaneswar, Chennai, Indore, Jabalpur, Jaipur, Kakinada, Kochi, Ludhiana, New Delhi Municipal Council, Pune, and Vishakhapatnam. The Indian Government, in consultation with respective state transport departments and urban local bodies (ULB), may issue orders for mandatory adoption of EVs in the aforementioned twelve cities.

Amongst key challenges, there is a consensus on the obvious higher costs for EVs and the need for EV charging infrastructure. In addition, the lack of consumer awareness and the need for EV and charging infrastructure technologies that are unique to the Indian electricity system and driving conditions are immediate impediments. Suggested action plan to these challenges are elaborated in the preceding sections.

A. *Suggested Action Plan*

Tables I, II, and III present summarized action plan for prioritized electric transportation sectors that have emerged from stakeholder workshops and this study's analyses. The three transportation sectors – public, government, and private – include 2-, and 3-wheelers, and light-, medium-, and heavy-duty EVs (e.g., cars, buses) categories. The ideal driving range per single full charge for all these categories is specified.

TABLE I. Public Transportation

	Public Transportation and Ideal Range per Full Charge [a]
1.	**Mini Buses (20-25 passenger capacity):** At least 50-kilometer (km.) or 31-mile (mi.) range per single charge.[b]
2.	**Taxis:** At least 100 km. or 62 mi. range per single charge.
3.	**Three Wheelers:** At least 100 km. or 62 mi. range per single charge.
4.	**Fleet-Service Vehicles, Chartered Buses, Vans, Shuttles, etc.:** At least 50 km. or 31 mi. range per single charge.
5.	**School Buses:** At least 50 km. or 31 mi. range per single charge.
6.	**Airport Buses:** At least 30 km. or 19 mi. range per single charge.
7.	**Passenger Buses** (50+ seating capacity)**:** At least 60 to 70 km. or 38 to 44 mi. range per single charge.[b]
8.	**Delivery Vehicles** (two-, three-, and four-wheelers, postal and courier vans, etc.)**:** At least 50 km. or 31 mi. range per single charge.

a. With optimized range for all of the public transportation, fast charging infrastructure along with alternate current (AC) slow charging with single phase or 3-phase in the public places will be required at airports, public parking slots, metro stations, government bhawans, bus depots, etc.

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[b] Generally, intra-city buses are classified into three main categories, based on the bus sizes (up to 8 meters or 26 feet, more than 8 meters but up to 10 meters or 33 feet, and more than 10 meters)

TABLE II. GOVERNMENT-OWNED VEHICLES

Government Vehicles and Ideal Range per Full Charge [c]	
1.	**Cars owned by the GoI and the state:** At least 50-60 km or 31-38 mi range per single charge.
2.	**Cars leased by the GoI and the state:** At least 50-60 km or 31-38 miles range per single charge.
3.	**PSU owned and hired cars and buses:** At least 100 km or 60 mi range per single charge
4.	**Non-combat vehicles or fleets of defence forces:** At least 50-60 km or 31-38 mi range per single charge

[c] Government vehicles under consideration are: The government of India (GoI), States, public sector undertakings (PSUs.), and the defense sector. The AC slow charging may be adequate for most government vehicles as trips are generally shorter and vehicles are parked for longer hours generally. These requirements may be further analyzed and fleet size may be split between slow charging and fast charging EVs to delineate EVs that support fast charging standards.

TABLE III. PRIVATE VEHICLES

Electrification and Ideal Range per Full Charge [d]	
1.	**Medium and heavy duty cars:** At least 100 -150 km or 60-90 mi range per single charge
2.	**Buses and shuttles:** At least 60-70 km or 38-42 mi range per single charge
3.	**Two wheelers:** At least 60-70 km or 38-42 mi range per single charge

[d] Considering the larger spatial coverage and diversity of drivers, both slow- and fast-charging requirements for private EVs are not analyzed in this study.

A typical bus in Delhi runs for about 205 km/day and operates for 50 to 60 km per single trip. This is the likely scenario for most of the buses in other major Indian cities [20]. To support this, the battery capacities must be analyzed for electric buses trips and must be designed for highest distance traveled with a single charge. Considering that the batteries are the highest cost in an EV, the right sizing of batteries (e.g., 60 to 70 km in Delhi) will lower the electric bus costs.

In each of the twelve lighthouse cities, above list of prioritized categories of vehicles may be mandated for electrification in a phased manner over the next five to seven years (long term sustained policy framework in support of NEMM). Each city may focus on different categories. For example, while Ludhiana may fast track 3-wheelers, Kochi may introduce mini buses. The action plan is intended to provide flexibility to each city to prioritize EV deployment based on their local needs. In an aggregate, all twelve cities should consider all categories for transportation electrification.

IV. ANALYSES OF EV AND CHARGING TECHNOLOGIES

Appropriate battery specifications, supporting electrical systems, and charging infrastructure for different types of vehicles for India are key for deployment of optimal technologies and leveraging the Smart Grid integration.

The battery guidelines can be used to develop appropriate specifications for battery sizes and electrical components based on the prioritized EV sectors and categories, determined earlier. The EV or automotive original equipment manufacturers (OEM) may design and size their cell modules and battery packs suitable for above ranges. If the cell modules and battery packs are inter-changeable amongst different makes of vehicles, the cost of batteries will be lower owing to larger volumes of similar designs. Also in such an environment battery swapping operations, especially for two-, and three-wheelers can be explored, that considers interoperability, safety and security of the vehicle equipment, drivers, and passengers. It is well understood that there are significant challenges with the battery-swapping solutions and requires a significant coordination and interoperability among batteries and EV OEMs, and other providers. However, this can be a longer-term strategy, while India explores immediate opportunity with installation of charging station infrastructure. Since the EV market in India is in its infancy, EV OEMs, battery OEMs, charging station OEMs, and technology integrators can make efforts to achieve standardization and interoperability among the EV systems, and electric grid as envisaged below.

1. All types of EVs can be charged from the types of charging stations that India will adopt.
2. Batteries of same sizes of vehicles can be swapped between – different 2-wheelers (e.g., scooters and motorcycles), 3-wheelers (rickshaws), and, potentially, light-and medium- duty vehicle types.
3. Efforts must be made to standardize the “under-the-hood” building blocks or cell-modules, so as to consolidate the market volume for “Make in India” effort through FAME, and to bring down the battery and component costs significantly.

All batteries of three- and four-wheelers may be capable of both alternate current (AC) and direct current (DC) charging. For the EV charging station, which are also technically termed and used interchangeably, as EV supply equipment (EVSE), and infrastructure, the ensuing requirements were analyzed and action plan for EV adoption were made.

For EVSEs, standards and interoperability are important, primarily due to different EV and EVSE OEMs, own-and-operate models, future proofing, and integration with electric grid systems. Such requirements and interoperability needs are well studied [21]. There are many standards development organizations (SDO) that develop power, safety, and communication standards for EVSEs. Among them, the Bureau of Indian Standards (BIS) International Electrotechnical Commission (IEC), International Standards Organization (ISO), Society of Automotive Engineers (SAE), and the Organization for Advancement of Structured Information Standards (OASIS) are the prominent ones across the world. The recommendations of standards or the organizations are not a comprehensive list, and more thorough analysis, specific to India, shall be made during later iterations.

The IEC 61851 is a key standard that addresses EV conductive charging system (connected power transfer between EV and EVSE) requirements for different AC and DC charging station equipment [22]. This standard has been taken as reference for the Automotive Industry Standard (AIS) 138 Electric Vehicle conductive AC charging system and the Part 1 draft is available for consideration in India [23]. Different EVSE charging classes or levels are based on *Maximum Load,* as defined in the AIS 138 and listed in Table IV. A near-term roadmap shall be defined later for a conductive DC charging system.

TABLE IV. CLASS OR LEVELS FOR ELECTRIC VEHCILE SUPPLY EQUIPMENT

Classes or Level for EVSE	
EVSE Class	***Maximum Load***
AC-Slow A	1 kW (1 phase)
AC-Slow B	2.2 kW (1 phase)
AC-Slow C	3.3 kW (1 phase)
AC-Fast A	12 kW (3 phase)
AC-Fast B	23 kW (3 phase)
AC-Fast C	45 kW (3 phase)

To ensure that appropriate charging infrastructure is available for different transportation sectors, specific recommendations were considered for different EV categories. These recommendations are listed in the following sections.

A. Two-Wheeler Charging Infrastructure

Normal AC slow charging is recommended, at least initially with socket/plug compatible with latest AIS standard (AIS 138, Part 3-DRAFT, August 2016) for charging infrastructure may be used for slow charging. Moving forward all new electric two wheelers may be equipped with appropriate types of batteries capable of AC fast charging. Based on the survey undertaken by Automotive Research Association of India (ARAI) for the power requirements, relative to EV charging requirements, AC slow charging (single phase) or AC fast charging (3-Phase) should be adequate for two wheelers.

B. Three- and Four-Wheeler Charging Infrastructure

The charging infrastructure for three- and four-wheelers such as rickshaws, cars, and mini-vans/buses includes recommendations for both AC and DC fast charging (DCFC) systems. An important consideration to alleviate perceived or real range anxiety and to encourage EV adoption among larger consumer groups, strategically-located DCFCs play a key role in reducing the time required for charging EVs.

In either of these cases various voltage, amperes, and power levels are recommended that are well suited for the Indian context and vehicle categories under consideration.

For three- and four-wheelers, the Level 1 charging would be using AC 230-volt, 15-ampere, which is about 3.45 kilowatt peak power, and for slow AC (e.g., residential) charging, industrial type 15-ampere plugs, socket-outlets, and couplers that are based on the IEC 60309 standard [24].

Level-2 (415-volt, ≤63-amperes) AC fast charging for up to 26 kilowatt using the IEC 62196 Type-2 connector standard is recommended. Requirements for specific inlet, connector, plug and socket-outlets for EVs are also referenced in the aforementioned AIS 138 standard. Additionally, to make the EVSEs compatible with global EV OEMs, support to SAE J1772 coupler standard may be considered.

For low-voltage DCFC, which are termed as <100-volt and up to 10 kilowatt maximum power level, an expression of interest (EoI) was published by the ARAI, on behalf of the Indian Ministry of Heavy Industry (MoHI), to develop, manufacture, deploy, and maintain 10,000 charging stations. The submitted proposals are under evaluation and the results will determine the next steps. These charging stations will be useful for 2-, 3- and light-duty 4-wheeler categories. Considering that this EoI prescribes that the charging stations support the Guobiao recommended (GB/T), or non-mandatory, DCFC connector standard, which is applicable for the Chinese EV and EVSE OEMs, India must also consider other globally adopted DCFC connector standards for low-voltage applications such as SAE J1772 combo-coupler standard (CCS) and IEEE 2030.1 standard [25] [26]. Both these standards are widely used in the North America, Europe, and Japan. Such support will encourage competition among EV OEMs that support these charging standards, provide customers choice of EVs, and lower the EV and grid-integration costs through global standards harmonization. For high voltage DCFC, options available for consideration are SAE CCS, IEEE 2030.1, and GB/T. The IEC 61851 standard Part 23 and 24 covers all these three types. It is also recommended that all three-wheelers, and potentially four-wheelers to be fitted with appropriate types of batteries capable of fast charging.

As a general requirement for all charging station levels, all charging outlets and charging stations must support a standard that enable ubiquitous and cost-effective communications between EV and EVSE, EVSE and third-party providers, and EV/EVSE with electric grid. The recommended standards for communication interfaces are:

- *Between EV and EVSE:* Recommend using IEC 61851. Additionally, ISO 15118 and Institute of Electrical and Electronics Engineers' (IEEE) 2030.5 must be reviewed [27] [28].
- *Between EVSE and third-party network (for EVSE remote management and payment systems):* Open Charge Point Protocol or OCPP.[1][29]
- *Between EVSE third-party network and utility or grid-operators:* OASIS Open Automated Demand Response (OpenADR 2.0) profiles and/or IEEE 2030.5. [30]

C. Large Buses Charging Infrastructure

While there is some standardization for light- and medium-duty EV charging infrastructure, unfortunately, there are no standards for heavy-duty EVs such as buses and trucks or how they are applicable to the Indian context (e.g., new distribution system lines along the transportation corridors). Various U.S and European research and standards organizations and EV/EVSE OEMs are reviewing standardization of charging infrastructure among one or more charging technologies and power standards. Since these standards for the buses and trucks are still under consideration, four charging technologies and power standards have been proposed by the SAE: (1) Manual conductive DCFC at high power based on SAE J1772 CCS [31], (2) Manual 3-phase conductive AC at high power based on SAE J3068 [32], (3) Overhead high-power conductive AC/DC based on SAE J3105 [33], and (4) Inductive (wireless) connection at high power based on SAE J2954 [34]. The use of charging infrastructure and standards should be based on analysis and applications in the Indian context.

In the United States and some European countries, Pantograph based overhead chargers are typically used for buses, which can be charged at a fixed route and stop. Ideal battery capacity can be considered based on earlier recommendations for fixed bus routes. With pantograph-based charging along the fixed bus route, battery size can be smaller, which will lower the cost of the electric buses and adoption costs. Also, smaller batteries will lower weight, increase the range, and accommodate more passengers. However, this requires a longer wait time of buses, which may not be ideal situation in the Indian contexts where roads are crowded and numbers of buses are higher. Under these circumstances, it is preferred to size the battery capacities appropriately and charge the buses at the end points (depots), as it can be managed safely in controlled premises. Some state transport agencies have shown preference to this approach. Here, the adoption of existing global standards for manual DCFC that are >300-volt to charge buses can be considered. The charging infrastructure can also be strategically located along the bus routes.

[1] The OCPP is a de-facto industry standard and is under consideration for formal standardization within the OASIS technical committee.

D. Charging Station Locations and Ownership Models

In addition to the charging infrastructure for different EV categories, the charging station location, ownership and operation models play a key role in the setting-up of the infrastructure. The following types of charging station types are recommended at strategic locations to promote EV adoption:

TABLE V. CHARGING STATION TYPES AND RECOMMENDED LOCATIONS

Charging Station Types and Recommended Locations	
Type of Chargers	***Recommended Locations***
AC Slow Charging	Residential Colonies, Bungalows, Commercial Centres/Buildings, Industrial Parks, University Campuses
Type 2 EVSE (AC Fast Charging - Refer AIS 138)	Bus Stands, Airports, Railway Stations, Sea Ports, Parking Lots , Malls, Movie Halls, Restaurants, Commercial Centres, Govt Offices , Industrial Parks, Hospitals/Medical Centres, Campuses, Hotels, Religious Places, Party Places/Marriage Halls
DCFC	Fuel Stations, Railway Stations, Airports, Sea Ports, Bus Stands, Parking Lots, Metro Stations, Malls, Campuses, Hospitals, Hotels, Strategic Locations on Highways
Bus Chargers (also for Trucks)	Bus Depots, Bus Routes
Two Wheeler Chargers	Homes, Offices, Industries and Shops

For each of these charging station types and recommended installation locations, the charging stations can be owned and operated by one or more of the following stakeholders: (1) DISCOMs; (2) parking lot and gas station owners; (3) building/campus owners/developers; (4) residential welfare associations and city governments; and (5) third parties.

Considering that these charging stations ownerhip and operation models, and their decoupling of EV driver ownerships, and integration with different Smart Grid domain and electricity markets (e.g., rate tariffs) necessiate the use of key standards for interoperabilty. Standards recommended earlier can future proof these charging stations assets, and enable innovation in technolgy and energy systems integration.

V. REVIEW OF ELECTRICITY PAYMENT MECHANISMS

The understanding and financial settlement of electricity used to charge the EVs is critically missing. This means, the charging and billing of services and the software that takes care of bill-to-cash management for EV charging or roaming is the key element. This process will ensure that the amount of electricity used by the customer will be invoiced accurately and in a cost-effective manner. For example, thorough review is needed, as to how in a multi-tenant building a charging infrastructure will be created and the customers invoiced for the electricity consumed. This process should use a proper system of customer identification and authorization. So all EV charging related details have to be collected and posted to contracts accounts of the DISCOMs. Customer inquiries also need to be managed. This process will ensure that there is no payment at charge station because it is too expensive and at the time of charging a valid contract (hand-shake between the EV and charging station should generate a contract between the EV customer and the DISCOM or their third-party franchisee) needs to be established. This handshake and identification needs standardization. The billing process should have a flexible pricing engine interface including prepaid and real-time pricing and payment options to encourage off-peak charging. More sophisticated solutions (e.g., via consumer interface apps) can allow for staggered/delayed charging as per customer requirements of ultimate charging time and quantum (e.g., full-charge by the morning). Also, capabilities related to location of all charging stations in the city should be available to EV drivers including real time status of charging, reservation of charging infrastructure, online/offline status, etc. Most of such features are currently deployed using the OCPP standard. The charging stations can also offer grid services to the DISCOMs, which can be compensated to the charging stations owners/operators and/or EV drivers.

VI. CONCLUDING RECOMMENDATIONS AND NEXT STEPS

Unlike India's pre-independence tryst, the present tryst with electric transportation represents opportunities for oil-independence and lowered carbon emissions, while benefiting its economic growth and improved quality of life to its people. Through stakeholder feedback and qualitative and quantitative review, this study reviewed key challenges and provided first-of-its-kind unique action plan for India to accelerate the near-term plans for EV adoption, and to meet the long-term NEMM and FAME goals. The recommendations also represent innovation opportunities for both transportation and electric industry. Some key concluding recommendations are:

- Request the Forum of Regulators (FOR) to issue amendments to the Indian Electricity Act so that third parties can set up charging stations, buy electricity from DISCOMs, and sell it to EV drivers at rate tariffs set by the State Electricity Regulatory Commissions (SERC).
- The ministries – Ministry of Power (MoP) and Ministry of New Renewable Energy (MNRE) enact new policy guidelines to lower entry cost barriers for EV adoption and accelerate customer acceptance of EVs. These guidelines must support EV infrastructure, grid- and consumer-friendly policies, rate tariffs, and deployment of second-life EV batteries, including its safe disposal.
- Further analyses are needed to assess the impacts of EV charging infrastructure on the electric grid and customer electricity costs, and propose standardized solutions to future-proof the EV charging assets.

A. Next Steps

The Indian MoP shall review these recommendations with ULBs and relevant state authorities from twelve cities, including respective DISCOMs, Central Electricity Authority, Central Energy Regulatory Commission (CERC), SERC, MoHI, MoUD, Department of Science and Technology, Bureau of Indian Standards, and industry stakeholders for another round of deliberations. The outcomes shall be used to assign responsibilities to different agencies with agreed upon action items to accelerate EV adoption. The following are the anticipated actions, as next steps:

- Conduct workshops with electric transportation stakeholders from twelve lighthouse cities to determine standards for AC, DCFC, and bus chargers; and also assign responsibilities to agencies to take lead and prepare project plans to establish charging infrastructure in these cities. This includes gathering of regulatory support from the CERC/FOR and public education.
- Prepare a roadmap for phased rollout of EV infrastructure in India, including potential for case studies to understand applications, specific to India.
- Advice key PSUs and large private companies to setup charging infrastructure at public places in the twelve

lighthouse cities. This can leapfrog the deployment. These organizations can also lead the way by transitioning their ICE vehicle fleet to EVs.

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